

Display Driver Configuration Guide for Windows CE.NET 4.2 MMSP™ 2

User Manual

MDMP2520FDisplayconfigUM

Ver. Preliminary / June 17,2005

MP2520F

Display driver configuration Guide for Microsoft

WindowsP®P CE.NET

MAGIC EYES

This document contains information on a product under development. MagicEyes reserves the right to change or discontinue this product without notice. © MagicEyes Digital Co.ltd 2005. All rights reserved.

Important Notice

Information in this document is provided solely to enable system and software implementers to use MagicEyes products. There are no express or implied copyright licenses granted hereunder to design or fabricate any integrated circuits or integrated circuits based on the information in this document.

MagicEyes reserves the right to make changes without further notice to any products herein. MagicEyes makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does MagicEyes assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation consequential or incidental damages. "Typical" parameters which may be provided in MagicEyes data sheets and/or specifications can and do vary in different applications and actual performance may vary over time. All operating parameters, including "Typicals" must be validated for each customer application by customer's technical experts. MagicEyes does not convey any license under its patent rights nor the rights of others. MagicEyes products are not designed, intended, or authorized for use as components in systems intended for surgical implant into the body, or other applications intended to support or sustain life, or for any other application in which the failure of the MagicEyes product could create a situation where personal injury or death may occur. Should Buyer purchase or use MagicEyes products for any such unintended or unauthorized application, Buyer shall indemnify and hold MagicEyes and its officers, employees, subsidiaries, affiliates, and distributors harmless against all claims, costs, damages, and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use, even if such claim alleges that MagicEyes was negligent regarding the design or manufacture of the part.

© MagicEyes Digital

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electric or mechanical, by photocopying, recording, or otherwise, without the prior written consent of MagicEyes Digital.

HOW TO REACH US:

MagicEyes Digital co., Ltd.

463-824 #501, A-ONE Plaza, 269-6 Seohyeon-Dong, Bundang-Gu
Seongnam-City, Gyeonggi-Do, Korea
TEL: 82-31-788-0300
FAX:82-31-707-5765

HOME PAGE:

HTU<http://www.mesdigital.com>TH

Developer Forum Site:

HTU<http://forum.mesdigital.com>TH

Table of Contents

1.	Introduction	4
2.	Configuration.....	4
2.1.	Example	4
2.2.	DisplayMode	5
2.3.	DisplayWidth	5
2.4.	DisplayHeight.....	5
2.5.	DisplayBpp.....	6
2.6.	Rotation.....	6
2.7.	VsyncPriority	6
2.8.	Gamma_R, Gamma_G, Gamma_B.....	6
2.9.	Brightness	6
2.10.	Contrast.....	6
2.11.	Saturation	7
2.12.	HUE	7
2.13.	UseALPHA	7
2.14.	ALPHA.....	7
2.15.	RGBOffatOverlay	7
2.16.	UseDithering.....	7
2.17.	Use2Daccel	7
2.18.	UsePostMemory.....	7I

1. Introduction

This document explains about configuration of display device driver included in Windows 4.2 BSP for MMSP2.

You can find the device driver in MMSP20\DRIVERS\DISPLAY folder.

2. Configuration

You can configure display device driver by changing or adding registry values

Parameters for configuration must be placed under

[HKEY_LOCAL_MACHINE\Drivers\Display\MMSP2] in registry.

2.1. Example

[HKEY_LOCAL_MACHINE\Drivers\Display\MMSP2]

"DisplayMode"=dword:1

"DisplayWidth"=dword:320

"DisplayHeight"=dword:240

"DisplayBpp"=dword:16

"Rotation"=dword:270

"VSyncPriority"=dword:14

"Gamma_R"=dword:64

"Gamma_G"=dword:64

"Gamma_B"=dword:64

"Brightness"=dword:80

"Contrast"=dword:1

"Saturation"=dword:48

"HUE"=dword:b4

"UseALPHA"=dword:0

"ALPHA"=dword:f

"RGBOffatOverlay"=dword:0

"UseDithering"=dword:1

"Use2DAccel"=dword:1

"UsePostMemory"=dword:0

2.2. DisplayMode

Default value : None.(depend on selected mode by the dip s/w setting)

It forced to set display mode as this value regardless of dip s/w setting.

The value is defined in Timings.h as MODENUMBER.

```
typedef enum
{
    ModeID_LCD_640x480x16_60Hz,           // 0
    ModeID_LCD_800x600x16_60Hz,           // 1
    ModeID_LCD_240x320x16_60Hz,           // 2
    ModeID_ARGB_640x480x16_60Hz,          // 3
    ModeID_ARGB_800x600x16_60Hz,          // 4
    ModeID_ARGB_854x480x16_60Hz,          // 5
    ModeID_DVI_800x600x16_60Hz,           // 6
    ModeID_TV_720x480Ix16_30Hz_SAA7120,   // 7
    ModeID_TV_720x480Ix16_30Hz,           // 8
    ModeID_TV_720x480Px16_60Hz,           // 9
    ModeID_TV_854x480Px16_60Hz,           // 10
    ModeID_LCD_800x480x16_60Hz,           // 11
    TOTAL_MODES
} MODENUMBER;
```

2.3. DisplayWidth

Default value : None (depend on selected mode by the dip s/w setting)

It forced to set display width as this value regardless of GpeMode.width in MMSP2DISMODE structure.

2.4. DisplayHeight

Default value : None (depend on selected mode by the dip s/w setting)

It forced to set display height as this value regardless of GpeMode.width in MMSP2DISMODE structure.

2.5. DisplayBpp

Default value : None (depend on selected mode by the dip s/w setting)
It forced to set display bpp as this value regardless of GpeMode.Bpp in MMSP2DISMODE structure.

2.6. Rotation

Default value : 270
Rotation can be enabled by USE_HWRGBROTATOR in BUILDOPT.h
The unit is degree.

2.7. VsyncPriority

Default value : 20
It is used to set a priority of the thread for VSYNC.

2.8. Gamma_R, Gamma_G, Gamma_B

Default value : 100
Adjust gamma correction. Range is from 1 to 1000 and the unit is one-hundredth. For example, value 100 means 1.0.

2.9. Brightness

Default value : 128
Adjust brightness. Range is from 0 to 255 and it is converted from -128 to +128. For example, value 128 means 0.0

2.10. Contrast

Default value : 1
Adjust contrast. Range is from 0 to 7 and it is converted from 0.0 to 1.875.
For example, value 1 means 0.125

2.11. Saturation

Default value : 72

Adjust saturation. Range is from 0 to 127 and it is converted from 0.0 to 2.0. For example, value 72 means 1.125.

2.12. HUE

Default value : 180

Adjust HUE. Range is from 0 to 360 and it is converted from -180 to +180. For example, value 180 means 0 degree. The unit is degree.

2.13. UseALPHA

Default value : 0

Enable or Disable alpha blending of overlay surface

2.14. ALPHA

Default value : 15

Set alpha value for alpha blending. Range is from 1 to 15.

2.15. RGBOffatOverlay

Default value : 0

Disable displaying RGB region that is overlapped with overlay region.

If set to 1, you can NOT see RGB at overlay region but it may improve video performance.

2.16. UseDithering

Default value : 1

Use 4x4 Dithering function of MLC

2.17. Use2Daccel

Default value : 1

Use 2D Accelerator for bit-blit function.

2.18. UsePostMemory

Default value : 0

Use Post-memory for Overlay, Set it as 1 if you wish to use scale-down function.